

Session Objectives

At the conclusion of this session, the student should be able to:

1. Recognize the potentially life-threatening dermatologic conditions both visually and through descriptions
2. Explain the pathogenesis of certain dermatologic conditions
3. Describe drug-induced, autoimmune or infectious etiologies of life-threatening dermatologic conditions
4. Manage best course of action for stabilization and treatment initiation for dermatologic emergencies

Session Outline

- Basics of Evaluation
- Urticaria/Angioedema
- Erythema Multiforme
- Stevens-Johnson Syndrome/Toxic Epidermal Necrolysis
- Staphylococcal Scalded Skin Syndrome
- Toxic Shock Syndrome
- Pemphigoid vulgaris
- Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS)
- Meningococemia
- Necrotizing Fasciitis
- Herpes Zoster Ophthalmicus
- Rocky Mountain Spotted Fever (RMSF)
- Burns
- Kawasaki Disease
- Measles

Basics of Evaluation

- Most rashes are not an emergency but it is important to identify if the rash is emergent
- **ABC's:** Airway, Breathing, Circulation

Keep an eye out for abnormal vital signs, O2 sat, mental status change

- **Thorough history:**

Events leading to pt's current presentation

Recent exposure to food, medications, plants, insects...

Patient allergies

Medications

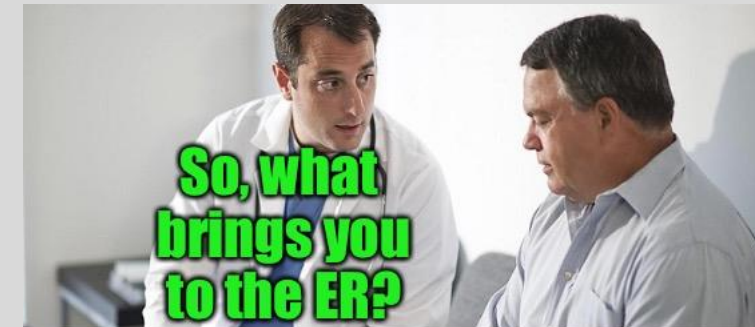
Past MHx

- **Physical exam:**

Appearance and distribution of lesions

Mucosal involvement?

Systemic signs



Angioedema

- **Management/Treatment:**

EARLY INTUBATION; Supply O2 to maintain >90% SpO2

Determine cause:

If due to **medication** (such as ACE inhibitor) **discontinue**

If **urticaria** also present, potential allergic trigger (meds, food, insect sting...)

Tx: Antihistamines, glucocorticoids, **epinephrine**

If **hereditary** variant,

Tx: **Icatibant** (a bradykinin inhibitor); **Ecallantide** (Kallikrein inhibitor); **C1 esterase inhibitor**; fresh frozen plasma if other tx not available

For pt with anaphylaxis: Prescribe autoinjector epinephrine

Identify trigger, avoid allergens, pt education

- **Disposition:** ICU

Erythema Multiforme (EM)

- **Cause:** 90% caused by infections; 10% caused by drug
Herpes Simplex Virus is #1 infectious cause (also *Mycoplasma pneumoniae*)
- **Pathogenesis:** EM develops 2/2 HSV infx via cell-mediated **immune** response directed at viral antigens deposited in lesioned skin
- **Diagnosis:** clinical; **targetoid lesions** with central dusky appearance
EM Minor: EM without (or only mild) mucosal disease and without associated systemic sx; resolves easily
EM Major: EM with severe mucosal involvement (may have systemic symptoms such as fever and arthralgias); conjunctivitis, cheilitis
- **Symptoms:** benign or malaise, low grade fever; **targetoid** skin eruptions develop over several days; may be pruritic or painful

Erythema Multiforme (EM)

- **Course/prognosis:**

Lesions appear over 3-5 days (soles and palms)

Resolves within 14 days

Usually don't scar, but can be hyperpigmented for months

Avoid recurrence usually by prophylaxis for HSV; if drug induced then avoid the drug

- **Management/Treatment:**

Treat underlying cause-

ex: antivirals (acyclovir) if HSV-related (and may be used to prevent recurrence)

Systemic steroids if severe

Dermatologist consult

Supportive care: fluid management, electrolyte, infectious, nutritional, thermoregulatory issues, pain control

Stevens-Johnson Syndrome (SJS) and Toxic Epidermal Necrolysis (TEN)

- **SJS/TEN:** epidermal detachment which is severe, life-threatening condition
 - SJS:** <10% of body surface area with epidermal detachment
 - TEN:** >30% of body surface area with epidermal detachment
- **NEED TO RECOGNIZE RIGHT AWAY TO KEEP PT ALIVE**
- **Cause:** typically, drug-induced; may have several doses of these meds before getting this reaction
 - Antibiotics: Sulfa meds and penicillin
 - Barbiturates
 - Anticonvulsants: lamotrigine, phenytoin, carbamazepine
 - NSAIDs
 - Allopurinol
- **Pathogenesis:** predominantly a drug-specific CD8+ cytotoxic T cell-mediated reaction → epidermal necrolysis
 - Potential genetic predisposition (HLA)
- **Diagnosis:** Clinical; + **Nikolsky sign** (the ability to extend the area of epidermal detachment with gentle lateral pressure); Skin biopsy shows keratinocyte necrosis

Stevens-Johnson Syndrome (SJS) and Toxic Epidermal Necrolysis (TEN)

- **Symptoms:**

About 1-3 weeks between first drug exposure and symptom onset

Prodrome: 1 -2 weeks of fever, arthralgia, anorexia before skin eruption

Rash: irregular shaped, dusky red, painful macules on trunk that quickly spread to upper extremities and face and coalesce; epidermis begins to slough off; progresses over hours to days; **mucous membrane involvement** (pain with swallowing)

- Pt is very aware- feels burning; tachycardic and hypertensive

Stevens-Johnson Syndrome (SJS) and Toxic Epidermal Necrolysis (TEN)

- **Prognosis:** dependent on % skin necrosis
SJS mortality rate: 5%
TEN mortality rate: 20+%
- At increased risk for electrolyte imbalance, multi-organ failure, infection, dehydration
- **Management/Treatment:**
Treated like a severe burn → ICU/burn unit care
D/C offending drug; future avoidance of all drugs that may cause SJS (2nd time will be worse)
Supportive care: correct electrolyte abnormalities, fluid replacement, pain control
If necessary: airway protection/ventilation; abx to avoid sepsis
- **Sequelae:** ocular (could lose vision); oral, pulmonary, dermatologic, genital (painful sex in future), psych (this is a pretty traumatizing disease)

Staphylococcal Scalded Skin Syndrome

- **Epidemiology:** Affects primarily children
- **Cause:** Skin barrier broken and leads to staphylococcus aureus infection, due to *S. aureus* exfoliative toxin
- **Diagnosis:** clinical; + Nikolsky sign
- **Symptoms:**
 - Fever, irritability, purulent conjunctivitis, rhinorrhea, tender skin
 - Skin becomes scalded, blistered, erythematous and edematous with flaccid bullae
 - Epidermis desquamates over 3-5 days leaving behind tender raw skin similar to burn
 - Mucous membranes spared**
- **Management/Treatment:**
 - IV antibiotics (nafcillin, oxacillin)
 - Supportive care: fluids, wound care, rehydration, thermoregulation, pain control

Toxic Shock Syndrome

- **Cause:** associated with tampon use; can be caused by other types of **packing**: wound, nasal, and abscess
- **Pathogenesis:** **Group A Streptococcus exotoxins (A, B, C)** or **Staphylococcus aureus (TSST-1)**; Superantigen-mediated T-cell activation → massive cytokine release → capillary leakage, **hypotension**, multiorgan failure
- **Prognosis:** Mortality rate of 6%, less with appropriate treatment
- **Diagnosis:** GAS or S. aureus, fever >102 F, hypotension, rash, desquamation, plus at least 2 of the following: diarrhea, vomiting, renal impairment, elevated liver enzymes, respiratory distress syndrome, disseminated intravascular coagulopathy, cutaneous erythema, soft tissue inflammation, or necrosis

Toxic Shock Syndrome

- **Symptoms:** fever >102 F, desquamative rash on the palms and soles, hypotension, multisystem organ involvement of three or more systems
 - 1-2 weeks after illness onset: pruritic maculopapular rash
 - 1-3 weeks after illness onset: desquamation, often of the palms and soles
- **Management/Treatment:**
 - Emergent removal of source of infection (i.e. remove the tampon/packing)
 - Supportive care: IV fluids, vasopressors if needed
 - IV antibiotics (clindamycin + vancomycin or nafcillin)
- **Prevention:** Proper tampon use, wound care, early treatment of infections with antibiotics

Pemphigus Vulgaris

- **Cause:** Autoimmune; IgG antibodies against desmoglein 3 leading to blistering eruption
- **Diagnosis:** + Nikolsky sign; skin biopsy shows IgG deposits in a reticular pattern along dermal epidermal junction on direct immunofluorescence
- **Symptoms:** flaccid bullae that rupture, leading to painful areas prone to secondary infx; mucosal involvement
- **Management/Treatment:**
 - Admit patient
 - Plasmapheresis and IVIG; Steroids and immunosuppressives
 - Treat fluid and electrolyte disturbances aggressively
 - Dermatology consult

Drug reaction with eosinophilia and systemic symptoms (DRESS)

- **Epidemiology:** 0.9 to 2 per 100,000 patients per year
- **Pathogenesis:** clear **drug** trigger in 80% of DRESS cases; associated with **HLA genes**; **Delayed Type IV** T cell-mediated hypersensitivity reaction → severe drug eruption, organ involvement and peripheral eosinophilia
- **Prognosis:** The mortality rate among patients with DRESS is estimated to be between 2 and 10 percent
- **Symptoms:**
 - The latency phase: time from drug initiation to onset of sx, 2-8 wks
 - Extensive skin rash, fever $\geq 101.3^{\circ}\text{F}$ and visceral organ involvement, lymphadenopathy, eosinophilia**
 - In contrast with SJS/TEN, mucosal involvement mild and skin detachment rarely seen
 - Liver injury** is the most common visceral manifestation
 - Facial edema
 - Can be mild, moderate or severe

Drug reaction with eosinophilia and systemic symptoms (DRESS)

- **Management/Treatment:**
 - Patients with severe DRESS: admit to hospital; organ support; ICU
 - Identify and withdraw causative medication
 - Avoid introducing new medications
 - Long taper of systemic corticosteroids
 - Supportive care includes fluid, electrolyte, and nutritional support
 - Gentle skin care, warm baths/wet dressings
 - Follow up with multiple specialists (ex: hepatologist, nephrologist, pulmonologist)

A 25-year-old patient presents to the emergency department with fever and a diffuse erythematous rash 10 days after being diagnosed with a UTI and beginning appropriate treatment. On physical exam, Nikolsky sign is positive. About 5% of the skin is involved, including mucous membranes. Which of the following is the most likely diagnosis?

- A) Erythema multiforme
- B) Staphylococcal scalded skin syndrome
- C) Stevens-Johnson Syndrome
- D) Toxic Shock Syndrome
- E) Toxic Epidermal Necrolysis

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- C) **Stevens-Johnson Syndrome**
- D) Toxic Shock Syndrome
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A 20-year-old female patient presents to the emergency department with fever, rash on her hands, and complaints of fatigue for one week. Her vitals include a temperature of 103 F and blood pressure of 80/50. On physical exam of the skin, a desquamating rash is present on the palms and soles. Which of the following will most likely also be a finding?

- A) Targetoid lesions on the trunk and extremities
- B) Painful subcutaneous nodules on the anterior lower legs
- C) History of tampon use
- D) Herpes Simplex Virus infection
- E) Recent antibiotic use

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- A) Targetoid lesions on the trunk and extremities
- B) Painful subcutaneous nodules on the anterior lower legs
- C) **History of tampon use**
- D) Herpes Simplex Virus infection
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A 55-year-old male patient presents to the emergency department with painful lesions on his trunk. On physical exam, + Nikolsky sign is present. A biopsy is performed and reveals IgG along the dermal-epidermal junction on direct immunofluorescence. Which of the following best describes the underlying pathogenesis of this patient's condition?

- A) Autoantibodies against desmoglein leading to disruption of keratinocyte adhesion
- B) Staphylococcal infection
- C) Cell-mediated immune response directed at viral antigens deposited in lesioned skin
- D) Recent anticonvulsant use

A 55-year-old male patient presents to the emergency department with **painful lesions** on his trunk. On physical exam, **+ Nikolsky sign** is present. A biopsy is performed and reveals **IgG along the dermal-epidermal junction on direct immunofluorescence**. Which of the following best describes the underlying pathogenesis of this patient's condition?

- A) **Autoantibodies against desmoglein leading to disruption of keratinocyte adhesion**
- B) Staphylococcal infection
- C) Cell-mediated immune response directed at viral antigens deposited in lesioned skin
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Meningococemia

- **Epidemiology:** most common in patients younger than 20 years
with most patients presenting younger than 5 years; Mortality 10-14%
- **Cause:** fatal systemic infection due to gram-negative diplococcus *Neisseria meningitides*; transmission via aerosolized droplets
- **Risk Factors:** asplenia; complement deficiency; crowding (prisons, dorms..)
- **Diagnosis:** lumbar puncture
- **Symptoms:** develop 3-4 days after exposure and rapidly progresses to severe illness
Headache, fever, altered mental status, nausea/vomiting, myalgia, arthralgia, neck stiffness, hypotension
Dermatologic findings: **petechial/purpuric rash**
- **Management/Treatment:**
Contact public health officials
Admit pt to ICU for close monitoring; Supportive care
Early antibiotic treatment w **IV third-generation cephalosporin + IV vancomycin**
ABC of resuscitation should be performed with IV fluid resuscitation for signs of shock
- **Prevention:** vaccination

Necrotizing Fasciitis

- **Epidemiology:** 0.4 per 100,000 people annually in the United States
- Aggressive, rapidly progressing skin and soft tissue infections → bacterial toxin release → muscle fascia and subcutaneous tissue necrosis and septic shock
- **Cause:**
can occur post-surgery, invasive procedures, or even minor procedure like phlebotomy, due to **breaking skin integrity**; most commonly gram-positive cocci *Staphylococcus aureus*, *Streptococci*, *Clostridium*, anaerobes
- **Risk Factors:** diabetes; immunosuppressed; trauma; liver cirrhosis; hx of alcoholism
- **Prognosis:** mortality rates range from 20 to 80%
Poor prognosis: certain streptococcal strains, advanced age, uncontrolled diabetes, immunosuppression, and delayed surgery

Necrotizing Fasciitis

- **Diagnosis:** must have high clinical suspicion when pt in intense pain, as infection may spread rapidly within hours
- **Signs and symptoms:**
 - excruciating pain out of proportion to presenting symptoms
 - systemic septic signs
 - tenderness to palpation beyond the erythematous border
 - cellulitis
 - bullae, ecchymosis of skin, paresthesia
 - Subcutaneous emphysema and crepitus
- **Management/Treatment:**
 - ASAP Surgery consult, needs immediate debridement; keep pt NPO
 - Ultimately transfer to ICU
 - Aggressive fluid resuscitation, pressors to counteract hypotension in septic shock
 - Early empiric broad spectrum antibacterial coverage

Herpes Zoster Ophthalmicus (HZO)

- **Symptoms:**

- Prodrome:

- vague symptoms; unilateral pain, photophobia, blurry vision, headache, and/or vision changes (no rash yet!)

- Active phase:

- 1-5 days after prodrome; skin lesions develop: **herpetiform cluster of erythematous vesicles**; evolve through stages of **pustulation, ulceration, and crusting**; lesions appear **unilaterally along the ophthalmic dermatome** and **do not cross the midline**

- Chronic Phase:

- can persist for months to years; chronic pain and hyperesthesia along affected dermatome

- Pain of HZO >30 days or after the resolution of the cutaneous lesions=
postherpetic neuralgia (PHN)

Rocky Mountain Spotted Fever (RMSF)

- Acute febrile tick-borne illness caused by *Rickettsia rickettsia*
- Transmitted by bite from American dog tick (*Dermacentor variabilis*), the Rocky Mountain wood tick (*Dermacentor andersoni*); endemic areas include **southeastern and south-central US**
- Mortality rate 20 to 30 percent without antibiotic treatment
- **Diagnosis:** high index of suspicion of RMSF in pt with flu-like sx during summer months, regardless of known tick exposure if they reside in or have history of travel to endemic areas
- **Pathogenesis:** infects vascular endothelial cells lining small and medium vessels → disseminated inflammation, loss of barrier function, and altered vascular permeability in body → systemic sx
- **Symptoms:** Begin 4-10 days after tick bite
 - Triad: **fever, HA, petechial or maculopapular rash starting at wrists and ankles**
 - Also, may present with myalgia, confusion, cardiovascular instability, lymphadenopathy

A 50-year-old female patient presents to the emergency room in excruciating pain. She states she recently returned from out of country where she had aesthetic surgery done. Her past medical history includes type 2 diabetes mellitus, and hypertension. On physical exam, there is tenderness to palpation on the left buttock and subcutaneous crepitus. What is the most appropriate next step?

- A) MRI
- B) IV antibiotics alone
- C) Surgical debridement as soon as possible
- D) High-dose steroids
- E) Skin biopsy

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- A) MRI
- B) IV antibiotics alone
- C) **Surgical debridement as soon as possible**
- D) High-dose steroids
- E) Skin biopsy

Kawasaki Disease

- **Epidemiology:** almost exclusively (90-95%) in **children** younger than 10 years; **Asian** population
- **Pathogenesis:** immune-mediated; inflammatory cells (macrophages, neutrophils, and lymphocytes) invade arteries (panarteritis) → disruption of internal elastic lamina of artery
- **Clinical Diagnosis: CRASH and Burn**
At least 4 of 5 criteria:
 - C**onjunctivitis
 - R**ash (palms and soles)
 - A**denopathy (cervical lymphadenopathy)
 - S**trawberry-appearing tongue with oral mucosal dry cracked lips
 - H**and and feet edema**And Burn: Fever > 5 days**
- **High risk of coronary artery vasculitis, including aneurysms and thrombosis if untreated**

Measles (Rubeola)

- **Symptoms:** fever and 4 “C’s”: cough, conjunctivitis, coryza, and “k”oplik spots
Three days after the onset of symptoms: maculopapular erythematous rash appears that starts on the face and spreads to the trunk and extremities

Complications include encephalitis, blindness, and subacute sclerosing panencephalitis (SSPE)- a late CNS degenerative disease occurring 5-15 years later

- **Management/Treatment:**

Contact public health officials

Supportive care is suggested for uncomplicated cases

Vitamin A: prevent corneal ulcerations and blindness, as well as reduce morbidity and mortality from measles

Prevention with MMR Vaccine

A 4-year-old male presents to the emergency room with 7 days of fever and hand and foot swelling. On physical exam, bilateral conjunctivitis, erythematous mouth, and rash of the palms and soles are present. After proper care is administered in the emergency department, which of the would be the best next step?

- A) Administer live attenuated vaccines within next two weeks
- B) Admit for treatment and echocardiogram
- C) Refrain from physical activity for a month
- D) Avoid NSAIDs for six months
- E) Expect symptoms to resolve within 48 hours

A **4-year-old** male presents to the emergency room with **7 days of fever and hand and foot swelling**. On physical exam, **bilateral conjunctivitis, erythematous mouth, and rash of the palms and soles** are present. After proper care is administered in the emergency department, which of the would be the best next step?

- A) Administer live attenuated vaccines within next two weeks
- B) **Admit for treatment and echocardiogram**
- C) Refrain from physical activity for a month
- D) Avoid NSAIDs for six months
- E) Expect symptoms to resolve within 48 hours

Summary

- Urticaria: first line, **epinephrine**
- Angioedema: **causes**: C1 esterase inhibitor deficiency; ACEI, NSAIDs
- Erythema Multiforme: **Herpes Simplex Virus** is #1; **targetoid** lesions
- Stevens-Johnson Syndrome/Toxic Epidermal Necrolysis: **drug** induced; **mucous membrane** involvement; must recognize right away
- Staphylococcal Scalded Skin Syndrome: Mucous membranes **spared****
- Toxic Shock Syndrome: packing; Group A Streptococcus exotoxins (A, B, C) or **Staphylococcus aureus** (TSST-1); **hypotension** and **shock**
- Pemphigoid vulgaris: **Autoimmune**; IgG antibodies against **desmoglein 3**
- Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS): **drug** induced; **HLA** genes; **eosinophilia**; liver injury
- Meningococcemia: Early antibiotic treatment w **IV third-generation cephalosporin and vancomycin**

Summary

- Necrotizing Fasciitis: breaking skin integrity; **excruciating pain**; signs of **sepsis**; **Subcutaneous emphysema** and crepitus
- Herpes Zoster Ophthalmicus: an **ophthalmic emergency**; latent **VZV** trigeminal ganglion reactivates, **ophthalmic branch of CN V** → **vesicular rash dermatomal distribution**, doesn't **cross midline**
- Rocky Mountain Spotted Fever (RMSF): **Rickettsia rickettsia** transmitted via American dog tick (**Dermacentor**); endemic areas **southeastern and south-central US**; rash starts **wrists** and **ankles**; tx with **doxycycline**
- Burns: **superficial, superficial and deep partial-thickness, full-thickness, deeper burn injury**
- Kawasaki Disease: immune mediated; **“CRASH and burn”**; High risk of **coronary artery vasculitis**, including aneurysms and thrombosis if untreated; tx: **IVIG and aspirin**; monitor for future cardiovascular complications
- Measles: single-stranded RNA paramyxovirus transmitted via respiratory droplets; **fever** and 4 **“C’s”**: **cough, conjunctivitis, coryza**, and **“k”oplik spots**; Tx **Vitamin A**